

Subgroups , Quotient groups

* Let G be a topological group and $H \subseteq G$.

Then H is a subgroup of topological group G iff H is a subgroup of group G and H is a closed set in topological space G .

Remark

H is a topological group equipped with induced topology on G . A subgroup of an abstract group which is a topological group is a topological group. This works as restriction of continuous maps is continuous.

* A normal subgroup N of topological group G if N is a normal subgroup of abstract group G .

Proposition

Let G be a topological group. Let H be a subgroup of abstract group G .

(i) $\overline{H}$ is a subgroup of top. group G

(ii) If H is normal subgroup of G then $\overline{H}$ is normal.

Proof:

(i) Observe $\overline{H} = \bigcap_{V \in \mathcal{V}} HV$ where $\mathcal{V}$ is collection of all nbd of e . As $H \subseteq \overline{H}$ hence $\overline{H} \neq \emptyset$.

For $x, y \in \overline{H}$ we want to show $xy^{-1} \in \overline{H}$.

We know $m: G \times G \longrightarrow G$ is continuous.
$$x \times y \longmapsto xy^{-1}$$

Observe $m^{-1}(\overline{H})$ is closed, and $H \times H \subseteq m^{-1}(\overline{H})$

Hence $\overline{H \times H} = \overline{H} \times \overline{H} \subseteq m^{-1}(\overline{H})$ and this implies

$$m(\overline{H} \times \overline{H}) \subseteq \overline{H}.$$

(ii) We construct a mapping

$$\begin{aligned} n: G \times G &\longrightarrow G \\ (x, y) &\longmapsto xyx^{-1} \end{aligned}$$

and observe n is a continuous map this implies

$n^{-1}(\overline{H})$ is closed and similar to (i)

$H \times H \subseteq n^{-1}(\overline{H})$ this implies $\overline{H \times H} = \overline{H} \times \overline{H}$

is a subset of $n^{-1}(\overline{H})$. Hence

$$n(\overline{H} \times \overline{H}) \subseteq \overline{H}.$$

□

* Set of all left cosets set G/H . We have a natural map or projection $\pi: G \longrightarrow G/H$.
$$x \longmapsto xH$$

Remark

If G is a topological group, we will topologize G/H assuming H is closed. Construct

$$\tau_{G/H} = \{O \subseteq G/H \mid \pi^{-1}(O) \in \tau_G\} \quad \text{i.e.}$$

O is open if and only if $\pi^{-1}(O)$ is open in G

which ensures π is a continuous map, moreover a quotient map where π is surjective by construction.

Lemma

G/H is a T_1 -space, and π is open.

Proof:

Since H is closed then πH is closed by homeomorphism. Observe $G - \pi H$ is open in G .

Let $\tilde{x} = \pi H \in G \setminus H$. Observe $\pi^{-1}(G \setminus H - \tilde{x}) = G - \pi H$. Hence $\tilde{x}$ is closed, $G \setminus H$ is T_1 .

We want to show π is an open map.

Let O be an open map in $G \setminus H$ then

$\pi(O)$ is open in $G \setminus H$ iff $\pi^{-1}(\pi(O))$ is open in G . Now $O \cap H = \pi^{-1}(\pi(O))$ and

$O \cap H = \bigcup_{h \in H} O \cap h$. Hence $O \cap H$ is open. So

$\pi^{-1}(\pi(O))$ is open which implies $\pi(O)$ is open.

lemma

G/H is a T_2 space.

Proof: Consider $xH \neq yH$ in G/H .

Now $x \in xH$ and $x \in G \setminus yH$ which is open as H is closed and yH is closed. Now, let V be a nbd of e st $x \in Vx \subseteq G \setminus yH$.

Observe $Vx \cap yH = \emptyset$ implies $VxH \cap yH = \emptyset$ otherwise $vxh_1 = gh_2$ and $vx = gh_2h_1^{-1}$ implying

$Vx \cap yH \neq \emptyset$. Now, we can find a nbd of e

V_1 st. $V_1^{-1}V_1 \subseteq V$ therefore $V_1^{-1}V_1xH \cap yH = \emptyset$

this implies $V_1xH \cap V_1yH = \emptyset$, which is

$\pi(V_1x) \cap \pi(V_1y) = \emptyset$ and by previous

lemma $\pi(V_1x)$ and $\pi(V_1y)$ are open sets

and $xH \in \pi(V_1x)$ and $yH \in \pi(V_1y)$. Hence

$G \setminus H$ is T_2



Remark

We call G/H the quotient space where H is a closed subgroup of G , has equipped with quotient topology.

lemma 7.

If G is a topological group, H is normal subgroup, the quotient topology equipped on the quotient group G/H is a topological group.

Proof:

We want to show $\Psi_1: G \setminus H \times G \setminus H \longrightarrow G \setminus H$
 $x_1, y_1 \longmapsto x_1 y_1^{-1}$
is continuous.

And $\Psi: G \times G \longrightarrow G$ is continuous
 $x, y \longmapsto xy^{-1}$

$$\begin{array}{ccc}
 G \times G & \xrightarrow{\pi \times \pi} & G/H \times G/H \\
 \pi \downarrow & & \downarrow \Psi_1 \\
 G & \xrightarrow{\Psi} & G/H
 \end{array}$$

Observe $\Psi_1(\pi \times \pi) = \Psi \pi$ and π, Ψ are continuous. Hence $\Psi_1(\pi \times \pi)$ are continuous.

Let O_1 be an open subset of G/H then

$\Psi_1(\pi \times \pi)^{-1}(O_1)$ is open. As $\pi \times \pi$ is an open map. ^{Observe} $\pi \times \pi(\pi \times \pi)^{-1}\Psi_1^{-1}(O_1)$ is open. Hence $\Psi_1^{-1}(O_1)$ is open.

□

Lemma

Let G be a topological group, and H be a subgroup

(a) If G is cpt then H and G/H are both cpt.

(b) If G is locally cpt then H and G/H are both locally cpt.

Proof:

(a) H is a closed subset of a cpt set and $G/H = \pi(G)$ where G is cpt and π is continuous mapping. Hence a continuous map of cpt set is cpt.

(b) A closed subset of a locally cpt set is locally cpt.

Let S be locally cpt, $T = \overline{T}$ and $T \subseteq S$.

For $p \in T \subseteq S$, there exists a nbd of p in S U_p st. $\overline{U_p}$ is cpt. Observe for $p \in T$

there exists an open set $T \cap U_p$ in T such that

$\overline{T \cap U_p} = T \cap \overline{U_p}$ is cpt as $T \cap \overline{U_p}$ is closed subset of $\overline{U_p}$ which is cpt.

Hence H is locally cpt.

Goal: G/H is locally cpt.

For $q_1 \in G/H$ and nbd of q_1 . Let
 $U = \pi^{-1}(U_1)$, and let $x \in G$ st. $\pi(x) = q_1$.
 As G is locally cpt we get a nbd of x O st.
 $\overline{O}$ is cpt and $\overline{O} \subseteq U$ by proposition given
 below. Then we have $\pi(\overline{O}) \subset \pi(U) = U_1$.
 If $O_1 = \pi(O)$ a nbd of q_1 . And $\pi(\overline{O})$
 is a continuous image of cpt set hence we get
 $\pi(\overline{O})$ is cpt. Since G/H is T_2 -space
 $\pi(\overline{O})$ is closed which implies
 $O \subseteq \overline{O} \Rightarrow O_1 \subseteq \pi(\overline{O})$ and
 $\overline{O_1} \subseteq \overline{\pi(\overline{O})} = \pi(\overline{O})$ which is cpt,
 so is $\overline{O_1}$. Thus O_1 is a nbd of q_1 such that
 $\overline{O_1}$ is cpt. It follows G/H is locally cpt.

Corollary

If G is locally cpt then G/H is normal or T_4 .

Proposition

For any topological group G , let U be a nbd of e , there exists a nbd of e V st. $\overline{V} \subseteq U$.

Proof.

By property 5) and 6) we know that for every nbd of e U we have a symmetric nbd of e st. $V^2 \subseteq U$. Now $x \in \overline{V} = \bigcap_{W \in \mathcal{V}} W$ where

$\mathcal{V}$ is set of all nbd of e , this implies $x \in \overline{V} \subseteq V^2 \subseteq U$.